Eosinophilic Esophagitis, Part One: Immunopathogenesis. Interleukin Thirteen and Eotaxin-3, Thymic Stromal Lymphopoietin and Basophils, a Leaky Barrier, Mast Cells and the Atopic March.

Imagine a nine-year-old boy who has had eczema since infancy. At the age of two he had anaphylaxis after eating peanut, and he carries an adrenaline auto-injector. Tonight he arrives in the emergency department because a piece of chicken is stuck in his oesophagus. At endoscopy the gastroenterologist removes the food bolus and sees an oesophagus with linear furrows, white specks and rings. The biopsies show sixty eosinophils per high-power field. His parents ask three sensible questions. Is this an allergy? Can skin tests or blood tests find the food that is causing it? And since he is so allergic anyway, would an anti-allergy injection, the kind used for asthma, fix it? The answers to all three questions depend on understanding how this disease works, and that is the subject of this chapter.

The main idea of the chapter is that eosinophilic oesophagitis is a chronic disease driven by food antigens through what immunologists call a type two immune response, and that the lining of the oesophagus, the epithelium, is not simply a victim of this response but its conductor. A single signalling molecule, interleukin thirteen, rewires the epithelial cells so that they call in eosinophils and, at the same time, lose the junctions that hold them together. That one idea explains the genetics of the disease, explains why tests based on the allergy antibody immunoglobulin E fail to find the trigger foods, and explains why every treatment we use actually works.

A few terms first. Eosinophils are white blood cells normally associated with allergy and parasites; healthy oesophageal lining contains essentially none. A high-power field is the area seen through the microscope at high magnification, and the diagnostic threshold for this disease is fifteen or more eosinophils in the most affected field. Type two inflammation is the branch of the immune system built around the signalling proteins interleukin four, interleukin five and interleukin thirteen, which are made by T helper type two cells and by group two innate lymphoid cells. Immunoglobulin E, usually shortened to I g E, is the antibody that causes immediate allergic reactions such as hives and anaphylaxis. Eotaxin-3 is a chemical attractant for eosinophils; its gene is called C C L twenty-six, and it acts on a receptor on eosinophils called C C R three. Thymic stromal lymphopoietin, usually shortened to T S L P, is an alarm signal released by stressed epithelial cells. Desmoglein-1 is a protein that forms the spot welds between skin-type epithelial cells. And calpain-14 is a protein-cutting enzyme found almost only in the oesophagus.

The evidence in this chapter comes from a series of laboratory and clinical studies. The epithelial gene signature and eotaxin-3 come from Blanchard and colleagues, published in the Journal of Clinical Investigation in 2006. The genetics come from genome-wide association studies by Rothenberg in 2010 and by Kottyan in 2014, both in Nature Genetics. The barrier story comes from Sherrill in 2014 and from later work on calpain-14 and a protective molecule called SPINK7. The role of T S L P and basophils comes from Noti and colleagues in Nature Medicine in 2013. The evidence that the disease is not driven by I g E comes from Clayton and colleagues in Gastroenterology in 2014. Twin and family data come from Alexander and colleagues in 2014, and the atopic march from a large birth cohort reported by Hill and colleagues in 2018. The link between untreated disease and strictures comes from a Swiss cohort reported by Schoepfer in 2013. Finally, the drug trials that test these ideas include the

dupilumab trials by Dellon in 2022 and by Chehade in 2024, the benralizumab trial reported by Rothenberg in 2024, and a phase three trial of cendakimab presented in 2024. The guidelines are the international AGREE consensus from 2018, the 2020 guideline from the American Gastroenterological Association and the Joint Task Force on allergy practice parameters, the 2024 paediatric guideline from the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, and the 2025 guideline from the American College of Gastroenterology.

Part one. One loop in five steps

It helps to picture the disease as a loop with five steps.

The first step is the barrier. In a person who will develop the disease, the oesophageal epithelium is a little leaky. It has less desmoglein-1 and less of the protective molecule SPINK7, and more calpain-14. Food proteins that would normally stay in the lumen can now slip between the cells.

The second step is the alarm. Stressed and leaky epithelial cells release alarm signals, chiefly T S L P and interleukin thirty-three. These signals attract and activate basophils and steer the local immune response towards type two inflammation.

The third step is the type two response itself. T helper type two cells and innate lymphoid cells make interleukin four, interleukin five and interleukin thirteen. Of these, interleukin thirteen is the master signal in the oesophagus.

The fourth step is recruitment. Interleukin thirteen acts on the epithelial cells and makes them secrete eotaxin-3. Eotaxin-3 draws eosinophils from the blood into the epithelium, and mast cells accumulate as well.

The fifth step is remodelling. Eosinophils and mast cells release transforming growth factor beta one, and the epithelium makes a matrix protein called periostin. Together these activate fibroblasts, lay down collagen beneath the epithelium, and make the oesophageal muscle stiff. Over years the oesophagus narrows.

Why call this a loop rather than a line? Because interleukin thirteen does not only attract eosinophils. In the same epithelial cells it lowers desmoglein-1 and raises calpain-14, which makes the barrier leakier still, so more food antigen enters and the cycle repeats. The good news is that breaking the loop at any point, by removing the antigen, by blocking interleukin thirteen signalling, or by calming the epithelium, resolves the inflammation.

Part two. The epithelial signature, with eotaxin-3 at the top

The modern understanding of the disease began with a 2006 study from Cincinnati by Blanchard and colleagues. They measured the activity of almost every gene in oesophageal biopsies from patients and controls.

Five hundred and seventy-four transcripts were abnormal in the disease, about one percent of the genome. What surprised the investigators was how uniform this signature was. It was almost identical in boys and girls, in young children and older patients, and in patients with and without other allergies. Only ten genes differed according to allergic status, one according to sex, and none according to age. A disease that looks the same at the gene level in a toddler and in a teenager, and in an atopic and a non-atopic patient, is one disease with one engine.

At the top of the list was eotaxin-3. Its messenger ribonucleic acid was on average fifty-three times higher in diseased oesophagus than in normal oesophagus. Eotaxin-3 levels tracked the number of eosinophils in the tissue very closely, with a squared correlation coefficient of about zero point eight four on microarray and zero point six five by the polymerase chain reaction method, which means that eotaxin-3 explained most of the variation in eosinophil count between patients. A common variant in the eotaxin-3 gene was more frequent in patients: children with two copies of the G form of this variant had an odds ratio for disease of about four and a half. And mice that lacked the eotaxin receptor, C C R three, were almost completely protected from oesophageal eosinophilia in an experimental model. Take away the receptor, and the oesophagus stays clear.

Part three. Interleukin thirteen, the master switch

If eotaxin-3 is the most visible product, interleukin thirteen is the switch that turns it on. When oesophageal epithelial cells are exposed to interleukin thirteen in the laboratory, they reproduce a large part of the disease signature. It is useful to walk through the main effects one at a time and link each to something we see in clinic.

First, interleukin thirteen recruits eosinophils by raising eotaxin-3 through a signalling protein inside the cell called STAT six. This produces the peak eosinophil count of fifteen or more per high-power field on which the diagnosis rests.

Second, interleukin thirteen weakens the junctions between cells by lowering desmoglein-1. Under the microscope this appears as dilated spaces between the epithelial cells, and functionally it means that antigens can enter more easily.

Third, interleukin thirteen raises calpain-14, a protease that is expressed almost only in the oesophagus. This helps explain why the disease affects the oesophagus rather than the rest of the gut.

Fourth, interleukin thirteen lowers SPINK7, an inhibitor that normally restrains protein-cutting enzymes in squamous epithelium. When SPINK7 falls, an enzyme called kallikrein-5 becomes more active, and the epithelium releases more T S L P. Barrier failure and inflammation then feed each other.

Fifth, interleukin thirteen raises periostin, which together with transforming growth factor beta one drives thickening of the basal layer of the epithelium, fibrosis beneath it and, eventually, strictures.

Finally, there is an important twist. Proton pump inhibitors, the acid-suppressing drugs, block the binding of STAT six to the eotaxin-3 gene in oesophageal cells, as shown by Cheng and colleagues in 2013. This effect does not depend on acid. It explains why about half of patients respond to a proton pump inhibitor even when they have no reflux at all, and why the old category of proton pump inhibitor responsive oesophageal eosinophilia has been folded into the disease itself.

The rule to remember is that every first-line therapy is an interleukin thirteen therapy in disguise. Diet removes the stimulus that triggers it. Steroids and proton pump inhibitors mute the epithelial response to it. And dupilumab blocks its receptor.

Part four. Two genetic loci explain the tissue and the allergy

Genome-wide association studies scan hundreds of thousands of common genetic variants to find those more frequent in patients than in controls. The 2014 study by Kottyan and colleagues combined seven hundred and thirty-six patients with nine thousand two hundred and forty-six controls, and it confirmed

two key regions.

The first is on the short arm of chromosome two, at a position called two p twenty-three, and it contains the gene for calpain-14. The risk variant roughly doubled the odds of disease, with an odds ratio of about two, and the result was extremely unlikely to be due to chance. Calpain-14 turned out to be the most highly expressed member of its enzyme family in the oesophagus among one hundred and thirty tissues surveyed. Interleukin thirteen raised its expression about four-fold in primary epithelial cells and more than one hundred-fold in a three-dimensional culture model. So this gene helps explain why the disease chooses the oesophagus.

The second region is on chromosome five, at five q twenty-two, and it contains the gene for T S L P. It was first identified by Rothenberg and colleagues in 2010 and was confirmed in the larger study, where the minor allele was protective with an odds ratio of about zero point seven three. Because T S L P is also involved in eczema and asthma, this locus helps explain why the disease travels with other allergic conditions.

Other variants with smaller effects lie in genes shared with atopic dermatitis and asthma, and there are rare inherited syndromes with high risk, including connective tissue disorders such as Ehlers-Danlos and Loeys-Dietz syndromes and the skin barrier disorder Netherton syndrome. These rare conditions point again to barrier biology and to transforming growth factor beta signalling.

The practical lesson is that neither calpain-14 nor T S L P variants are diagnostic tests. They are common in the population, each adds only modest risk, and together they explain only part of who gets the disease.

Part five. Genes or home?

If genes explain only part of the risk, what explains the rest? The most informative study is the twin and family registry from Cincinnati, reported by Alexander and colleagues in 2014.

In identical twins, if one twin had the disease, the other had it in about fifty-eight percent of cases. In non-identical twins the figure was about thirty-six percent. That difference was not statistically significant, which already suggests that genes are not the whole story. When the investigators modelled the data properly, separating genes from the shared family environment, the heritability, meaning the share of variation explained by genes, was only about fourteen and a half percent, while the common family environment explained about eighty-one percent. If the shared environment was ignored, as in a simple family analysis, heritability appeared to be about seventy-two percent. In other words, earlier estimates were inflated because families share kitchens, antibiotics, birth seasons and microbes as well as genes.

The family risk is nevertheless real. About two point four percent of siblings and one point eight percent of first-degree relatives also had the disease, which is between ten and sixty-four times the rate in the general population. The risk was strongly skewed towards males: the recurrence risk ratio was about sixty-four for brothers compared with sisters. Among early-life factors, a larger difference in birth weight between the twins and birth in the autumn were associated with one twin having the disease and the other not.

So when a parent asks whether a younger brother is at risk, the honest answer is yes, considerably more than other children, and that most of that risk seems to come from the shared environment rather than

from inherited genes alone.

Part six. A leaky epithelium is both cause and consequence

Three barrier molecules deserve a closer look, because each is moved in the wrong direction by interleukin thirteen.

Desmoglein-1 is a cadherin, a kind of molecular velcro, that sits in desmosomes, the spot welds between cells. In active disease its level falls. Sherrill and colleagues showed in 2014 that loss of desmoglein-1 splits epithelial cells apart, weakens the barrier, and itself increases periostin, so barrier loss feeds remodelling. When the disease goes into remission, desmoglein-1 returns.

Calpain-14 is the oesophagus-specific protease we met in the genetics. When it is over-expressed experimentally, it strips desmoglein-1 from the cell junctions and weakens the barrier, which links the risk gene directly to the leak.

SPINK7 is a serine protease inhibitor that normally protects squamous epithelium. Its loss, described by Azouz and colleagues in 2018, unleashes the enzyme kallikrein-5, and the result is more T S L P and more inflammation.

The visible consequence of all this on biopsy is dilated intercellular spaces, which are scored in the eosinophilic oesophagitis histology scoring system, together with basal zone hyperplasia, the thickening of the deepest layer of the epithelium. These features track the barrier and improve with treatment.

The practical importance is threefold. The barrier helps explain why atopic dermatitis, a skin barrier disease, so often comes first. It raises the possibility that sensitisation to foods happens through damaged skin before the gut ever meets them. And it reminds us to look beyond the eosinophil count when judging response on biopsy.

Part seven. An allergy that does not need I g E

Many children with this disease also have classic I g E-mediated food allergy, like the boy in our case. It would be natural to assume that the oesophageal disease is simply another I g E reaction. The evidence says otherwise.

In 2013 Noti and colleagues built a mouse model of food-induced oesophagitis that developed even in mice that could not make I g E. The disease did, however, require T S L P and basophils. Blocking T S L P or removing basophils both prevented the disease and treated it once it was established. In human biopsies the same group found more T S L P and more activated basophils, and patients with a gain-of-function variant of the T S L P gene had stronger basophil responses.

In 2014 Clayton and colleagues looked at the human side from another angle. They found deposits of immunoglobulin G four, a different antibody class, and many immunoglobulin G four producing plasma cells in the oesophageal tissue of adults with the disease, and high levels of food-specific immunoglobulin G four in their blood. In the same work, a randomised trial of omalizumab, the antibody that removes I g E, reduced neither oesophageal eosinophils nor symptoms compared with placebo. The 2025 American College of Gastroenterology guideline now advises against omalizumab for this disease.

Why does this matter? Because it explains why skin prick tests, specific I g E blood tests and patch tests do not identify the foods that drive the oesophageal disease. All recent guidelines advise against choosing an elimination diet on the basis of these tests. Instead, foods are removed empirically, with milk as the most common trigger, and the effect is checked with repeat biopsies.

Part eight. The last stop on the allergic march

The atopic march describes the typical order in which allergic diseases appear in childhood: eczema in infancy, then food allergy, then asthma and allergic rhinitis. Hill and colleagues asked where eosinophilic oesophagitis fits, using a primary care birth cohort of one hundred and thirty thousand four hundred and thirty-five children from Philadelphia. One hundred and thirty-nine developed the disease, a prevalence of about one in a thousand.

The peak age at diagnosis was a third of a year for eczema, one year for I g E-mediated food allergy, just over one year for asthma, two point one years for allergic rhinitis, and two point six years for eosinophilic oesophagitis. So the oesophageal disease arrives last.

Each prior allergic condition raised the rate of a later diagnosis. The strongest association was with I g E-mediated food allergy, with a hazard ratio of about nine, meaning nine times the rate of later diagnosis. Atopic dermatitis carried a hazard ratio of about three, allergic rhinitis about two point eight and asthma about one point nine, and each additional allergic condition raised the rate further. All of these associations were statistically robust.

The interpretation is that the oesophageal disease is the late step of the allergic march, after the skin, the food and the airway. That food allergy carries the greatest risk fits the barrier-first, antigen-driven story. And the practical consequence is that a young child with eczema and food allergy who refuses food, vomits or chokes deserves an endoscopy with biopsies rather than a label of picky eating or reflux.

Part nine. Eosinophils are not alone

The disease is named after the eosinophil, but several other cells matter.

Eosinophils are recruited by eotaxin-3 through the C C R three receptor and depend on interleukin five for survival. They release toxic granule proteins such as major basic protein and eosinophil peroxidase, and they release transforming growth factor beta one. They are the histological marker of the disease. Mast cells accumulate inside the epithelium under the influence of interleukin thirteen, interleukin nine and stem cell factor. They release tryptase and transforming growth factor beta one and are thought to contribute to muscle dysfunction and symptoms. Basophils, activated by T S L P, release interleukin four and amplify the response without I g E. T helper type two cells and group two innate lymphoid cells, driven by T S L P, interleukin thirty-three and interleukin twenty-five, produce the upstream signals: interleukin four, interleukin five and interleukin thirteen. Finally, fibroblasts and smooth muscle cells respond to transforming growth factor beta one and periostin by laying down collagen and stiffening the wall.

The most striking evidence that eosinophils are not the whole disease comes from the MESSINA trial of benralizumab, published in 2024. Benralizumab binds the interleukin five receptor on eosinophils and destroys them. In two hundred and eleven patients, eighty-seven percent of those on benralizumab reached a histological response of six or fewer eosinophils per high-power field, compared with six and a half percent on placebo. Yet difficulty swallowing did not improve more than with placebo. The

eosinophils were gone, but the disease was still there. The lesson is that eosinophils mark the disease, while the epithelium, the mast cells and fibrosis carry much of the symptom burden.

Part ten. Time turns inflammation into stricture

What happens if the loop is never broken? A Swiss cohort of two hundred adults, reported by Schoepfer and colleagues in 2013, answered this by relating the prevalence of strictures at diagnosis to how long symptoms had gone undiagnosed.

Among patients diagnosed within two years of symptom onset, about seventeen percent already had strictures. Among those whose diagnosis had been delayed by more than twenty years, about seventy-one percent had strictures. The relationship was time-dependent and statistically very strong.

The mechanism is the remodelling step of the loop: transforming growth factor beta one and periostin activate fibroblasts, and epithelial cells themselves can take on features of connective tissue cells, a process called epithelial to mesenchymal transition. This explains a well-known clinical pattern. Children usually have an inflammatory phenotype, while adults more often have a fibrostenotic phenotype with rings and narrowing. These are not two diseases but one disease seen at two points in time. It also explains why symptoms change with age, from feeding refusal in infants, to vomiting, to abdominal pain, and finally to dysphagia and food impaction in older children and adults. The paediatrician's task is to stop the clock before collagen is laid down.

Part eleven. What the biologic trials taught us

Clinical trials of targeted antibodies have acted as experiments in human pathogenesis.

Dupilumab blocks the alpha chain of the interleukin four receptor, which is shared by interleukin four and interleukin thirteen, so it shuts down both signals at once. In adolescents and adults, in the trial reported by Dellon in 2022, weekly dupilumab produced histological remission, defined as six or fewer eosinophils per high-power field, in sixty percent of patients compared with five percent on placebo, and it also improved difficulty swallowing. In children aged one to eleven, in the EoE KIDS trial reported by Chehade in 2024, the higher-exposure regimen produced histological remission at sixteen weeks in sixty-eight percent of children compared with three percent on placebo, with the lower-exposure regimen reaching fifty-eight percent. Dupilumab is now approved for children from one year of age who weigh at least fifteen kilograms.

Benralizumab, as we saw, removed eosinophils without improving swallowing. Cendakimab, an antibody against interleukin thirteen itself, produced histological remission in about twenty-nine percent of patients compared with about two percent on placebo at twenty-four weeks in its phase three trial, with a small reduction in days with difficulty swallowing. Earlier, mepolizumab, which neutralises interleukin five, also lowered eosinophils without a clear benefit on symptoms.

The interpretation is that blocking the upstream type two signal at the epithelium heals both the tissue and the swallow, while deleting the downstream eosinophil does not. Mechanism predicted the trials.

Part twelve. Where each therapy cuts the loop

We can now map the treatments onto the loop. A dietary elimination removes the antigen at the barrier; milk is the most common trigger, which is why stepwise diets usually start with milk. A proton pump inhibitor blocks STAT six at the eotaxin-3 gene and may protect the barrier, independent of its effect on

acid. A swallowed topical steroid broadly suppresses the type two epithelial signature. Dupilumab blocks the receptor shared by interleukin four and interleukin thirteen. And dilation treats the fibrotic narrowing only; it does nothing for the inflammation, which continues if it is not treated.

So choosing between diet, a proton pump inhibitor and a topical steroid is really a choice of where to interrupt the same loop, made together with the family according to their goals, the child's diet and the phenotype. Dilation can relieve a narrowing, but it never replaces anti-inflammatory therapy.

Part thirteen. What the guidelines say

How do the main documents compare on the issues raised by pathogenesis? They are aligned on the concept of the disease. The 2018 AGREE consensus defines it as a chronic, immune and antigen-mediated oesophageal disease; the 2020 American Gastroenterological Association and Joint Task Force guideline and the 2024 European paediatric guideline share this view; and the 2025 American College of Gastroenterology guideline describes it as a type two, allergen-induced disease and states that it is no longer seen as a consequence of acid reflux. They are aligned in no longer requiring a trial of a proton pump inhibitor before diagnosis, a change made by AGREE in 2018. They are aligned in advising against elimination diets based on allergy tests. And they are aligned against anti-I g E therapy, with the 2020 guideline restricting it to clinical trials and the 2025 guideline advising against it.

There is a material difference on blocking the interleukin four receptor. The 2020 guideline, written before the trial data existed, restricted such biologics to research. The 2025 American College of Gastroenterology guideline recommends dupilumab for patients aged twelve and over who do not respond to a proton pump inhibitor, and suggests it for children aged one to eleven. The European paediatric guideline sits between them, presenting it as an option in refractory disease. Guidance also differs on fibrosis as an endpoint: the 2025 guideline asks that the endoscopic reference score be recorded at every endoscopy, while older documents say less.

The material gap is that no guideline yet uses molecular markers, the barrier or mast cells to choose therapy for an individual patient. The pathway is understood better than it is used.

Part fourteen. Common traps

Several errors come directly from misreading the pathogenesis. The first is ordering skin prick or specific I g E tests to choose foods for elimination, when they do not predict the trigger. The second is reaching for anti-I g E therapy because a child is atopic, when the oesophageal disease is not driven by I g E. The third is calling a response to a proton pump inhibitor reflux, when the drug acts on eotaxin-3 and the barrier independent of acid. The fourth is looking only at the eosinophil count on follow-up biopsies, when basal zone hyperplasia, dilated intercellular spaces and fibrosis beneath the epithelium also tell us about response. The fifth is equating depletion of eosinophils with cure, which the benralizumab trial disproved. The sixth is dismissing family history, when siblings and especially brothers carry a much higher risk. The seventh is waiting for dysphagia in a toddler, whose feeding refusal and vomiting are the same disease at an earlier stage. The eighth is dilating a stricture without treating the inflammation, so that the stricture returns. And the ninth is expecting a single gene test to make the diagnosis.

Part fifteen. What to do differently in clinic

Let us return to the nine-year-old boy with eczema, peanut anaphylaxis and a food impaction. Is this an allergy? Yes, in the sense that it is a food-antigen-driven type two disease, and it sits at the end of the allergic march that began with his eczema and his peanut allergy. But it is not an I g E allergy in the way his peanut reaction is. That answers the second question: skin and blood allergy tests will not find the foods that drive his oesophagus, and an elimination diet, if the family chooses one, should be empirical, usually starting with milk, and checked with a repeat biopsy after eight to twelve weeks. It also answers the third question: an anti-I g E injection is not the right tool and did not work in trials. If he needs a biologic, it would be dupilumab, which blocks the shared receptor for interleukin four and interleukin thirteen, the upstream switch of the disease. His first-line options remain a proton pump inhibitor, which works on eotaxin-3 rather than through acid alone, a swallowed topical steroid, or a diet. And because untreated inflammation turns into fibrosis over years, the goal is to break the loop now, in childhood, rather than to dilate strictures in his thirties. If he has a younger brother who refuses solids or vomits, that brother deserves a low threshold for endoscopy.

The broader changes for clinic are these. Explain the disease to families as an allergic disease of the oesophageal lining that does not run on the usual allergy antibody. Do not use I g E-based tests to choose foods. Think of every therapy as cutting the same interleukin thirteen loop at a different point. Look at the whole biopsy, not just the eosinophil count. Ask about siblings, especially brothers. And treat early, because time turns inflammation into stricture.